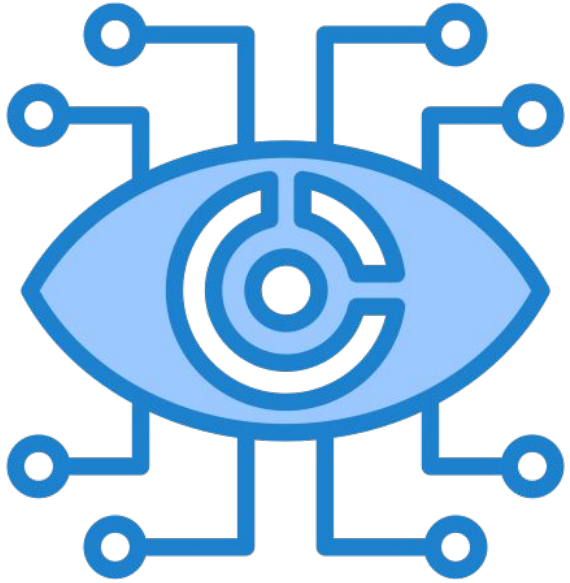
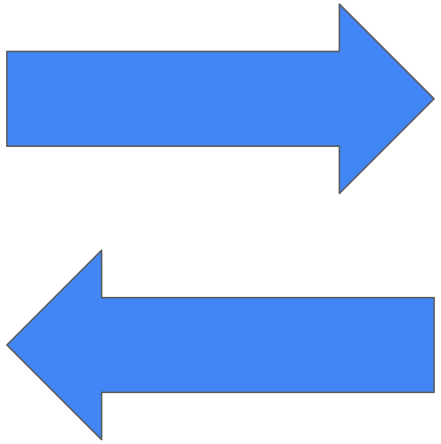
IMAGE CAPTION GENERATOR

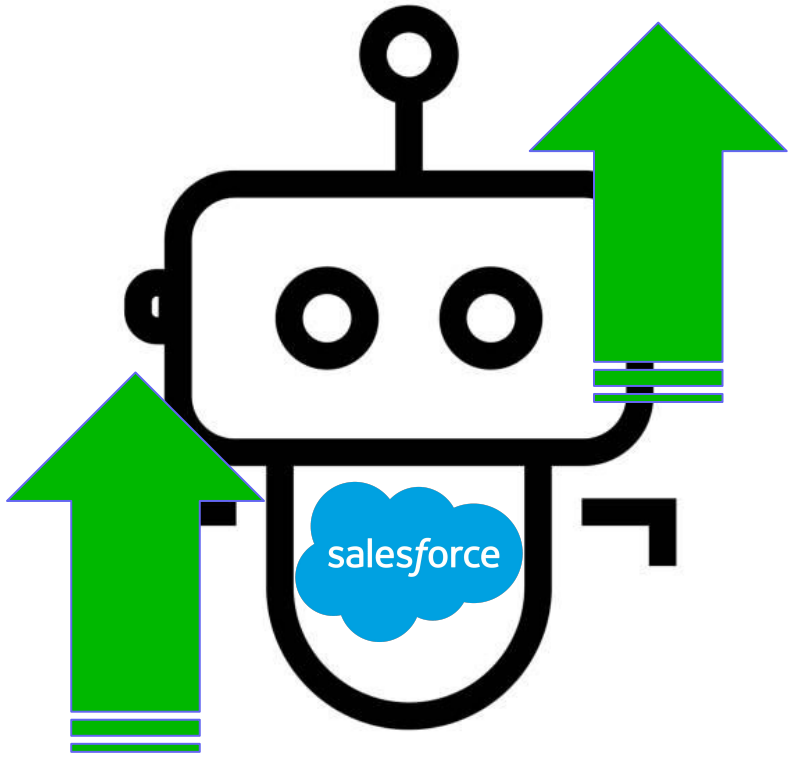
Sam Yuen
Ragul Magesh
Nithin Raja
Carina Davis

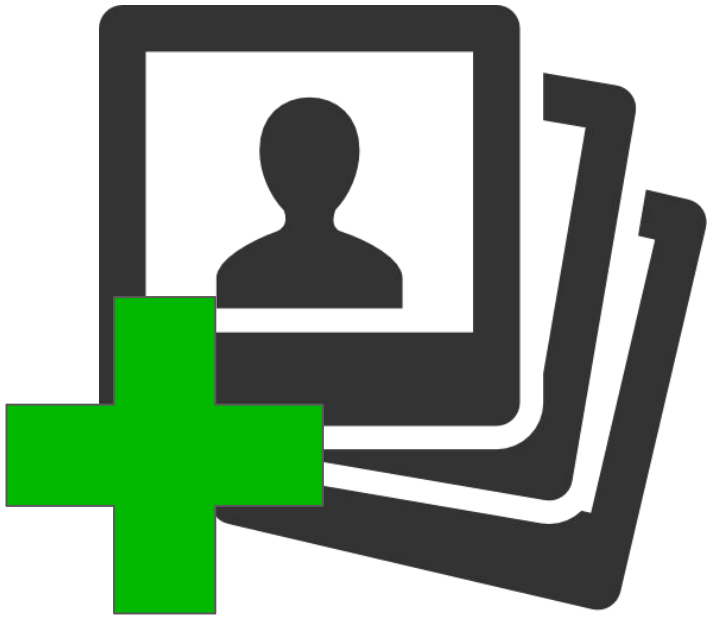
OBJECTIVES

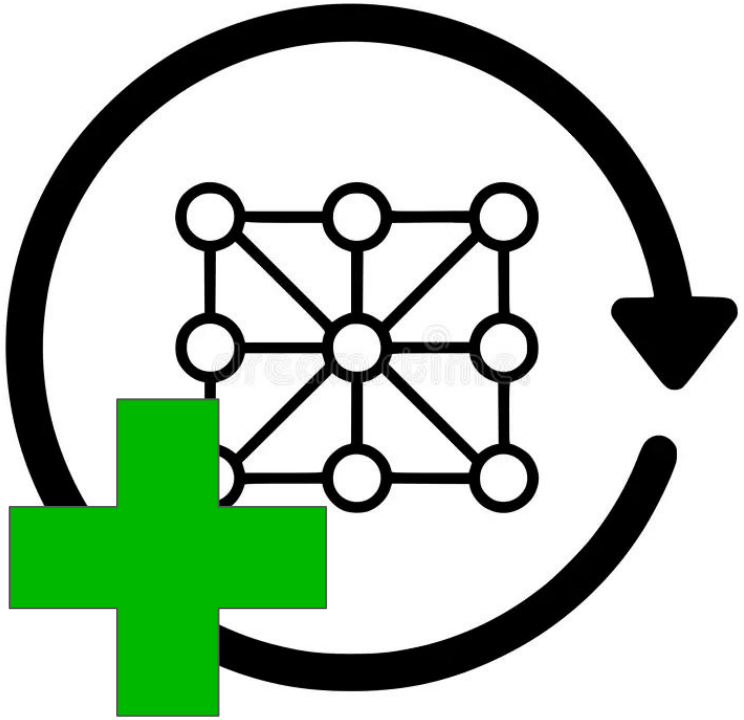




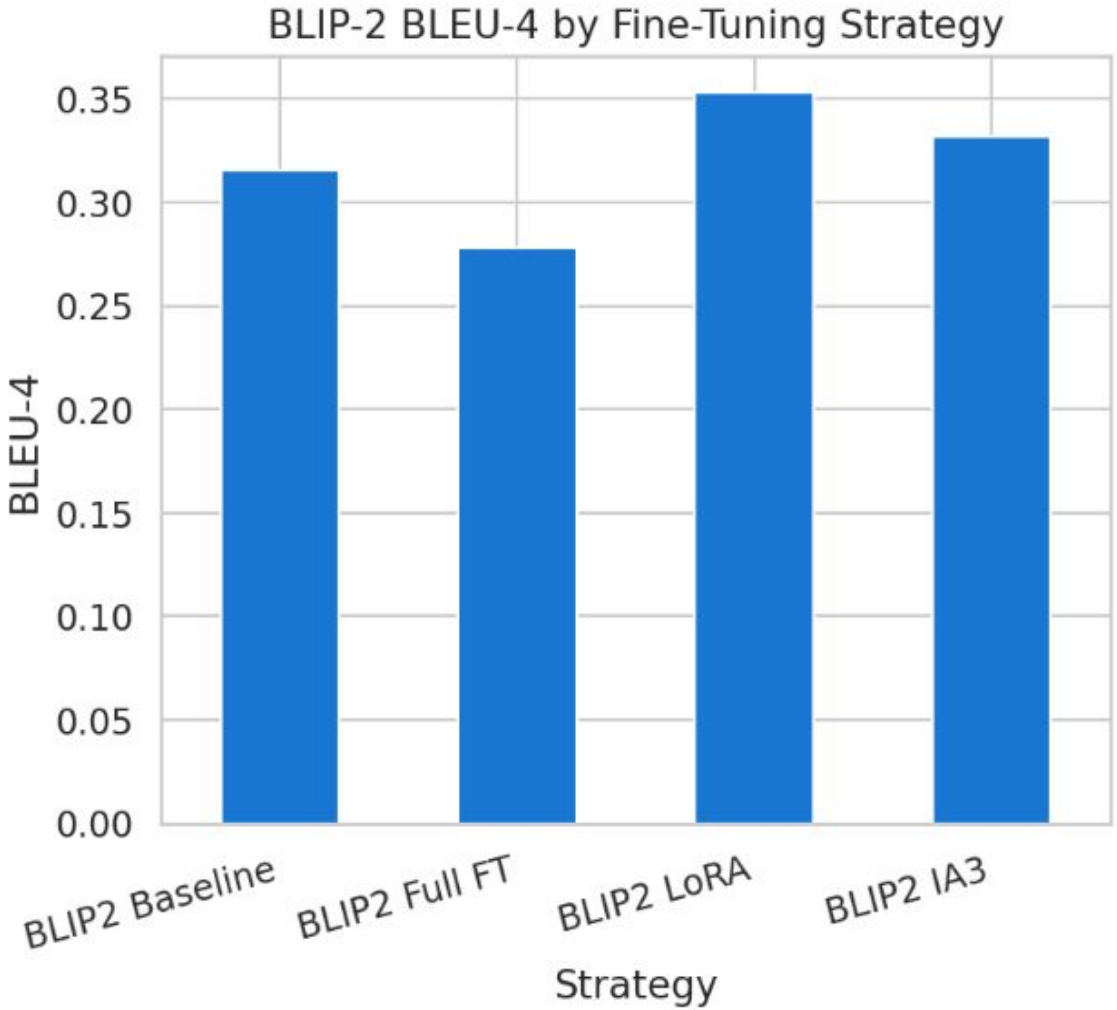
APPROACH

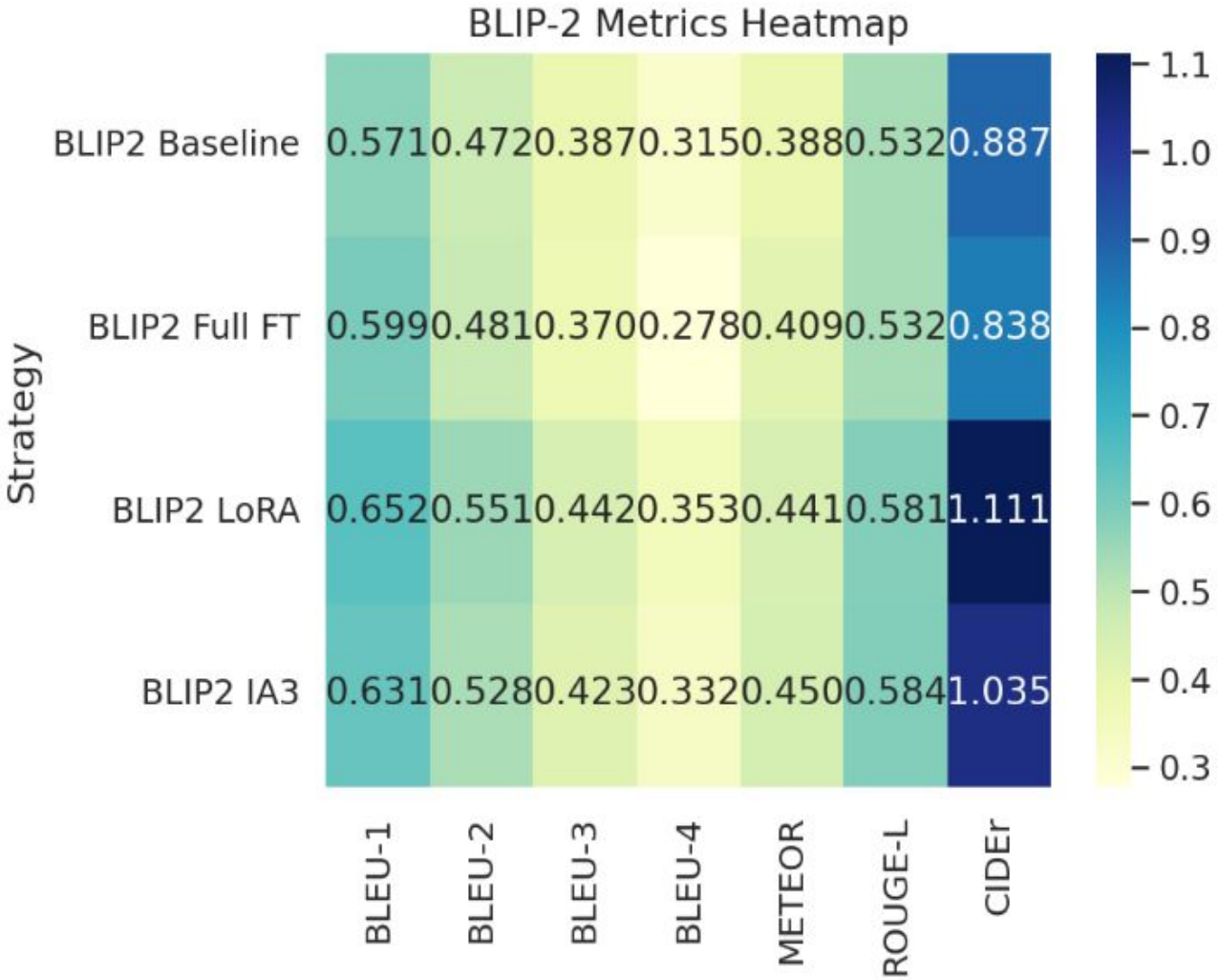




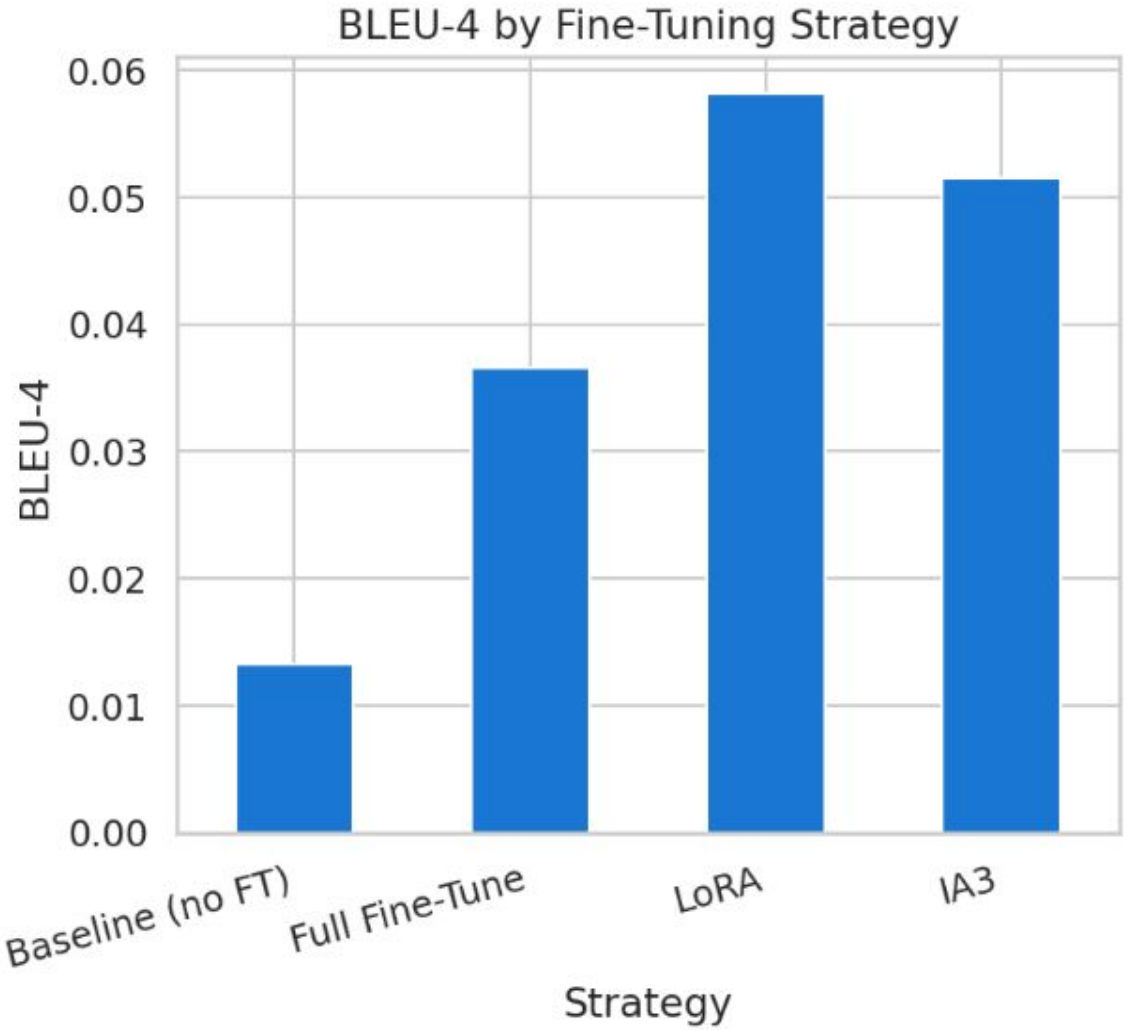


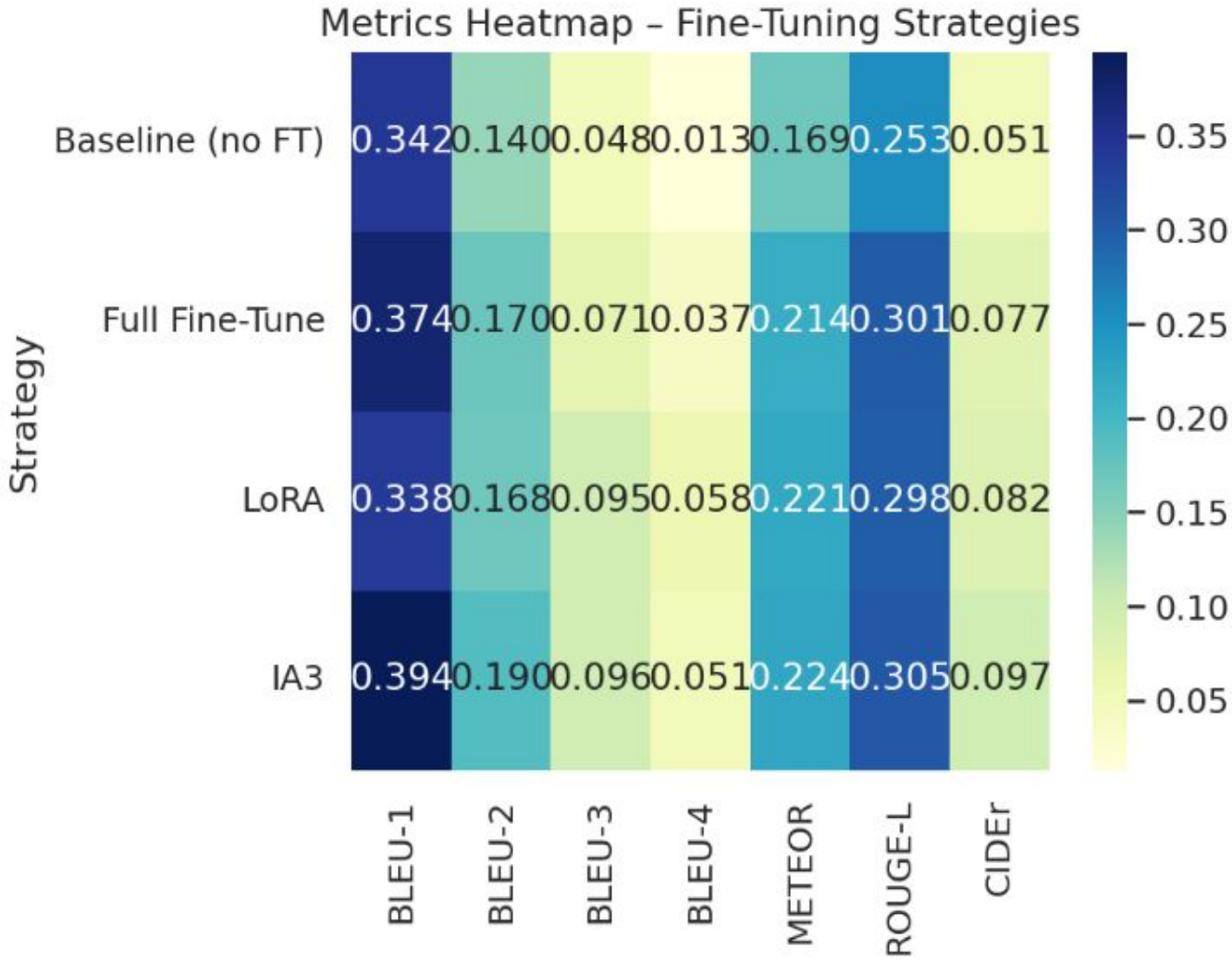
BLIP → BLIP 2


BLIP-2 BLEU-4 by Fine-Tuning Strategy
BLEU-4
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
BLIP2 Baseline
BLIP2 Full FT
BLIP2 LoRA
BLIP2 IA3
Strategy

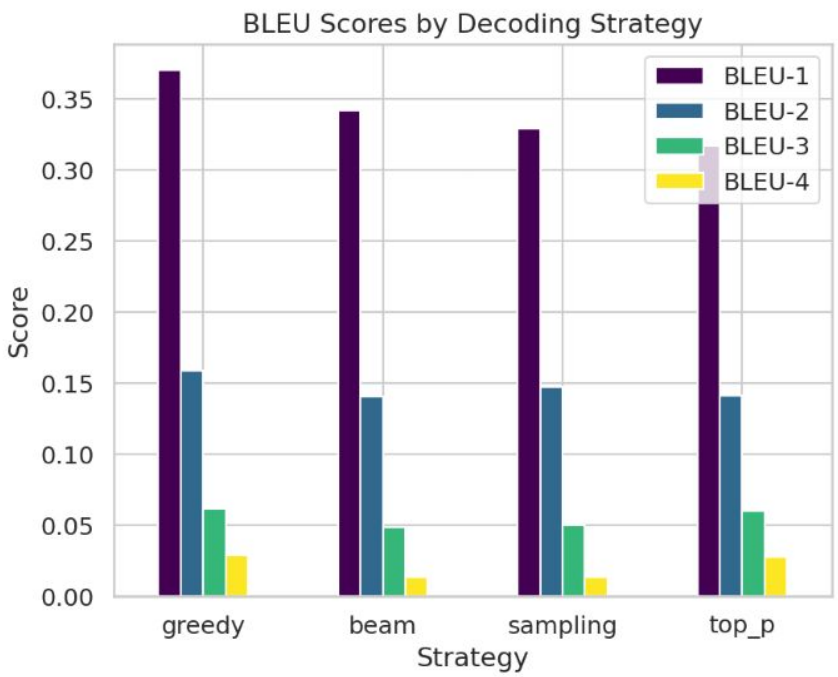

BLIP-2 Metrics Heatmap
Strategy
BLIP2 Baseline 0.571 0.472 0.387 0.315 0.388 0.532 0.887
BLIP2 Full FT 0.599 0.481 0.370 0.278 0.409 0.532 0.838
BLIP2 LoRA 0.652 0.551 0.442 0.353 0.441 0.581 1.111
BLIP2 IA3 0.631 0.528 0.423 0.332 0.450 0.584 1.035
BLEU-1
BLEU-2
BLEU-3
BLEU-4
METEOR
ROUGE-L
CIDEr
1.1
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3

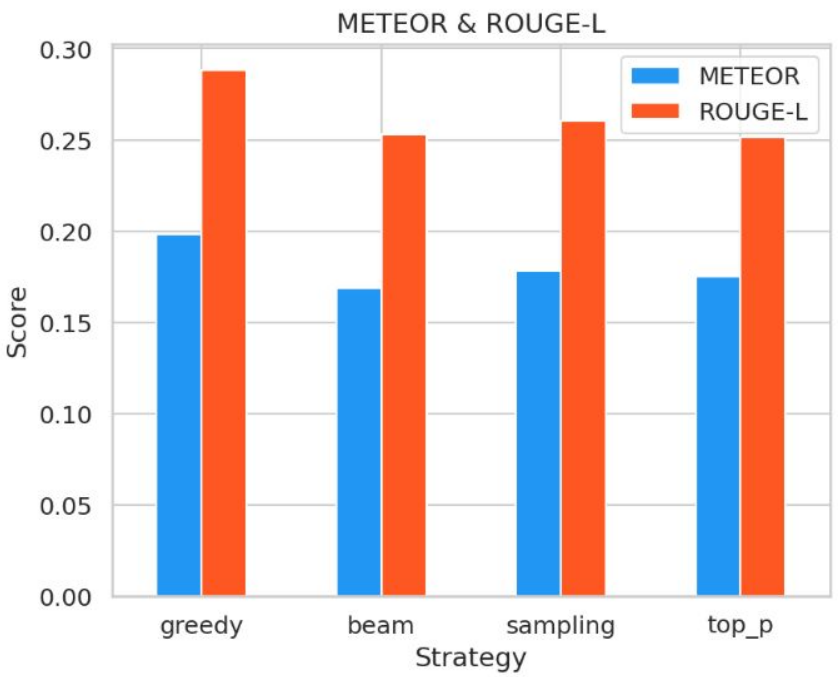
3k → 10k IMAGES

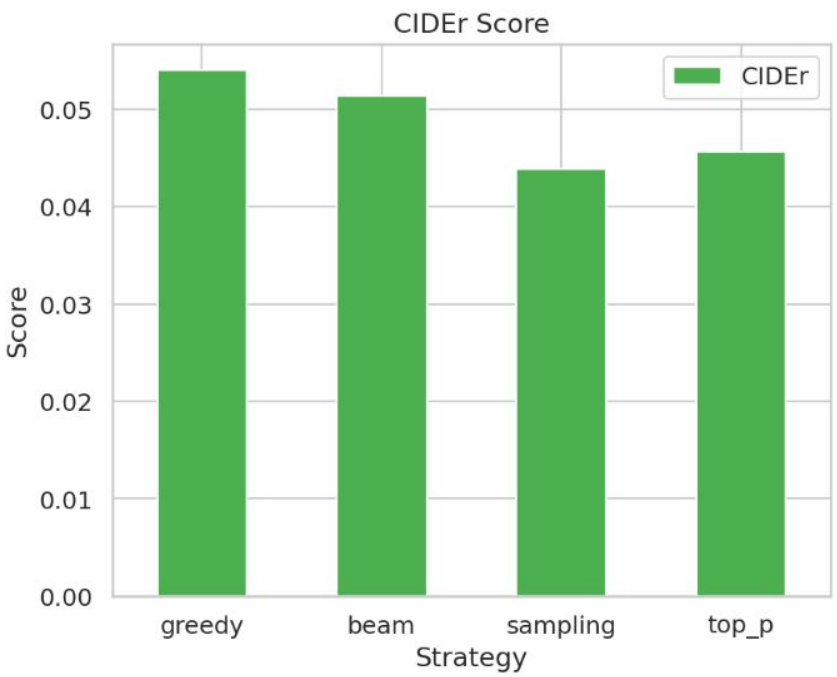




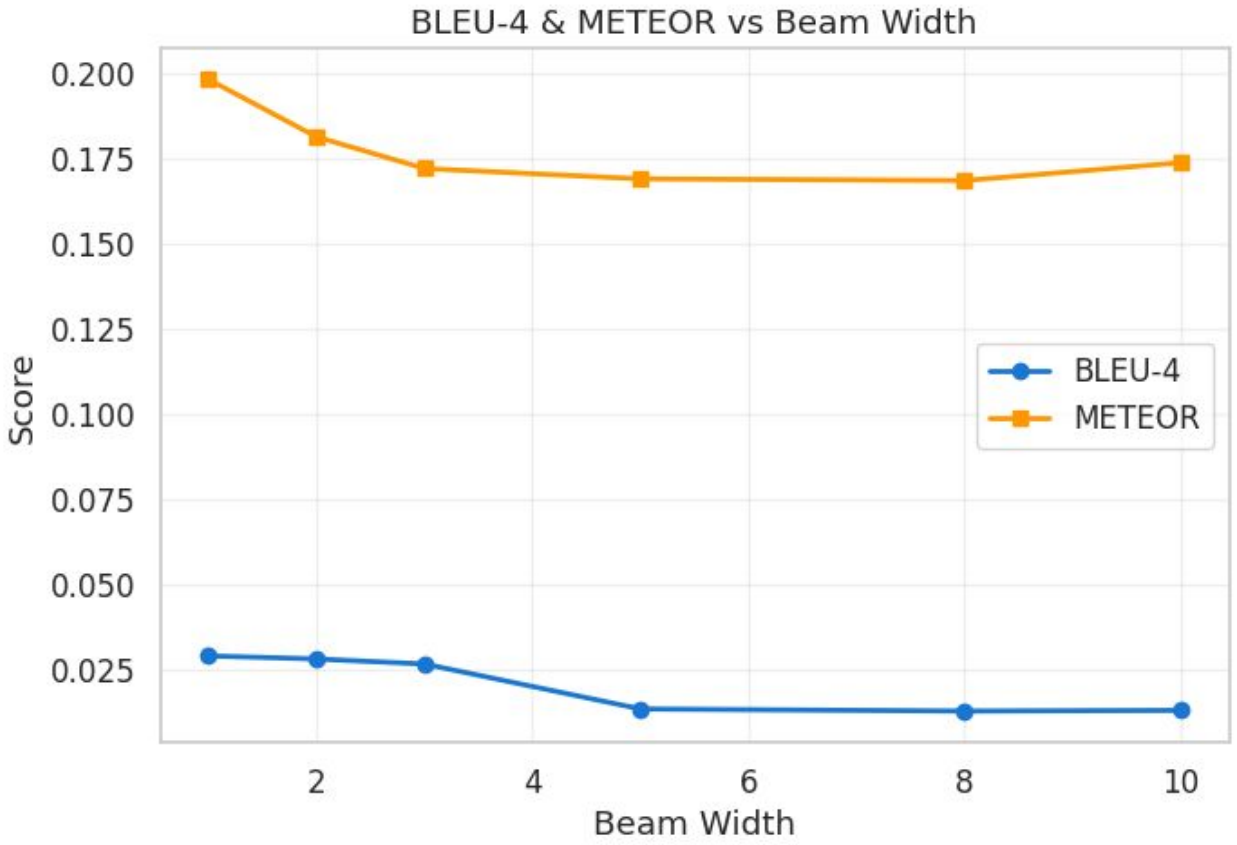
3k → 10k IMAGES

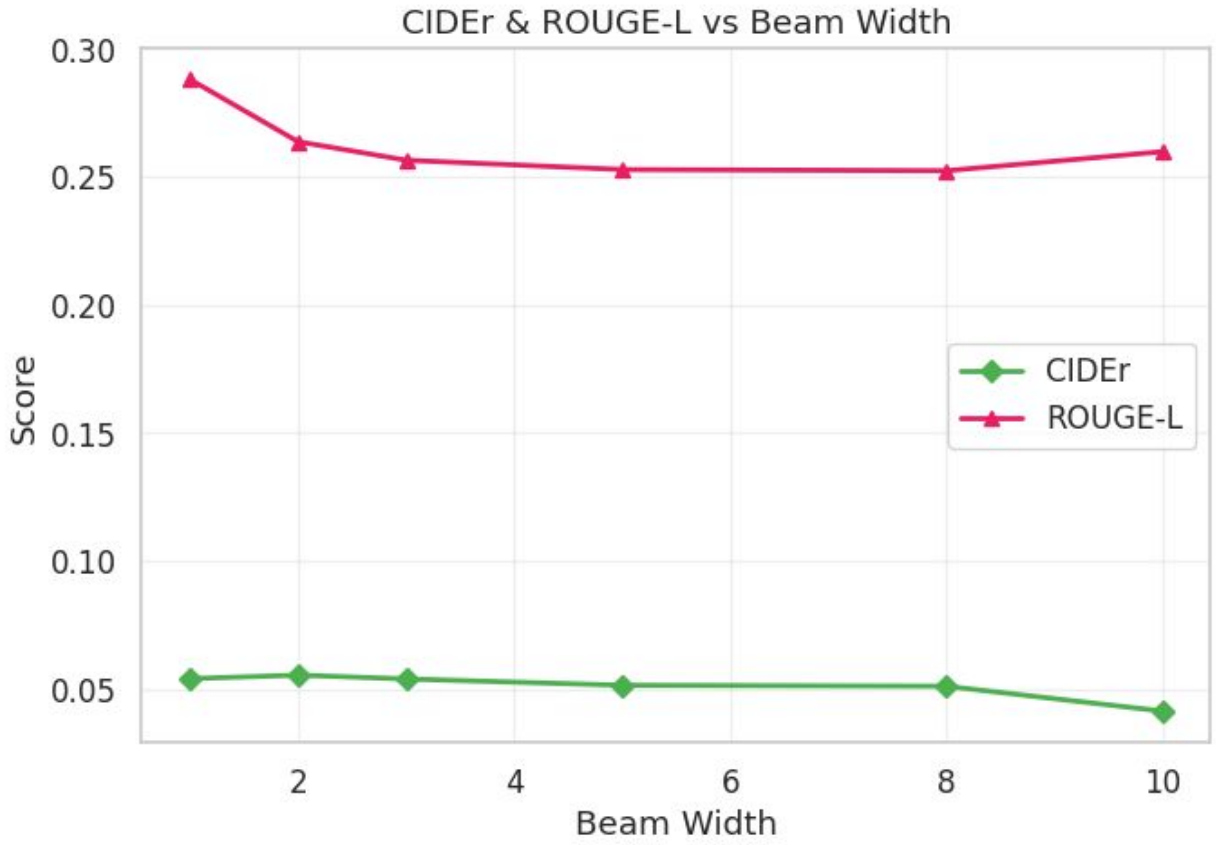




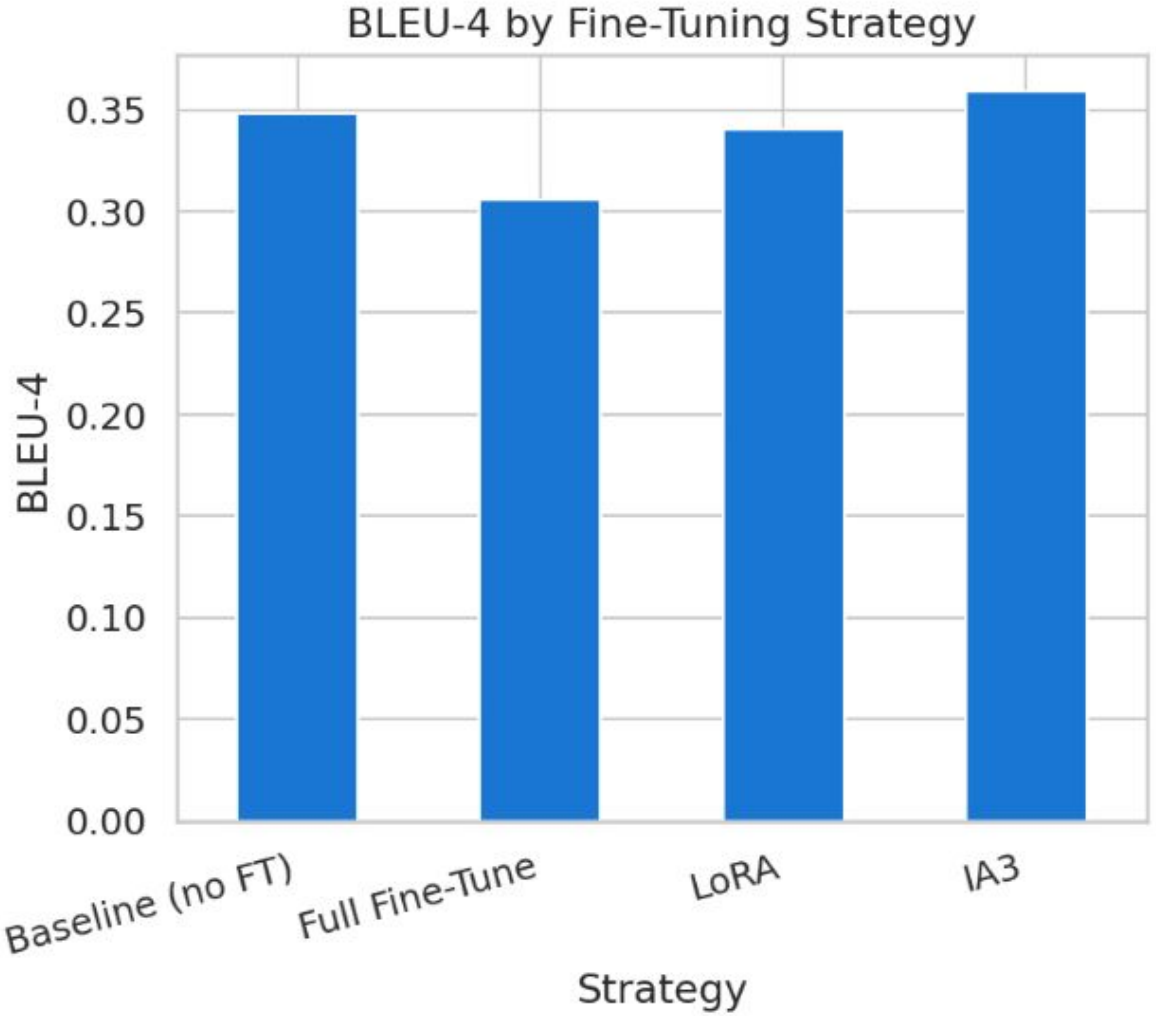


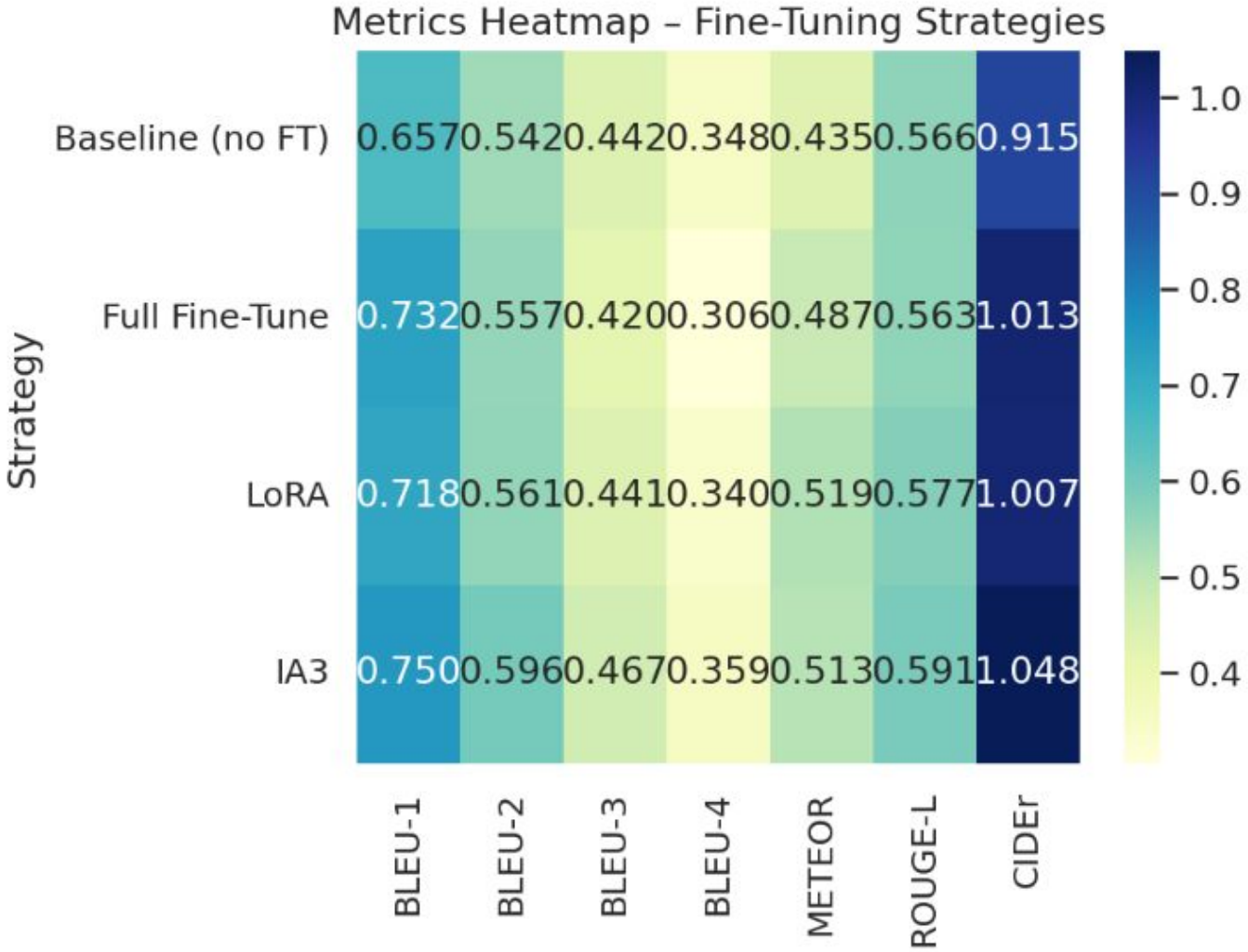
3k → 10k IMAGES



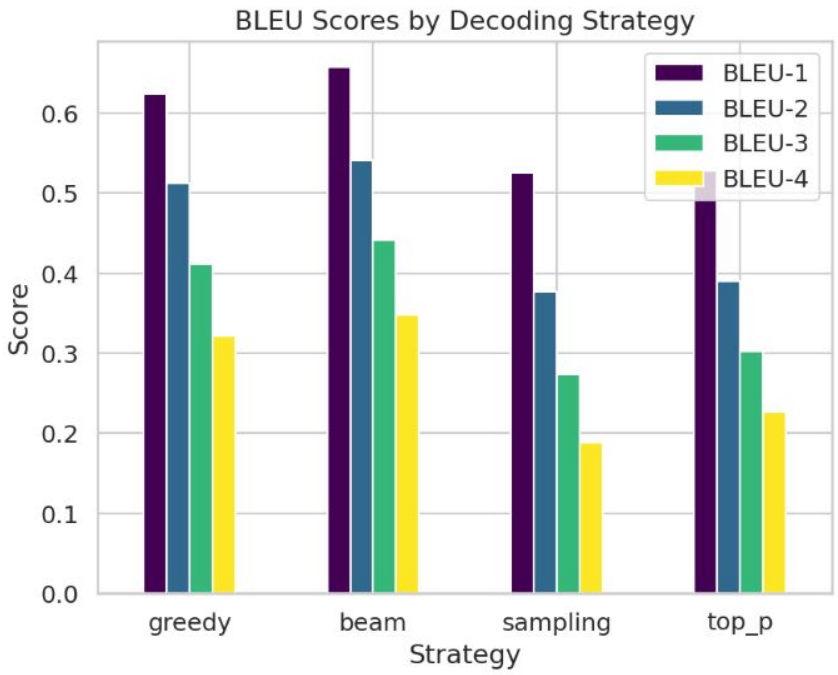


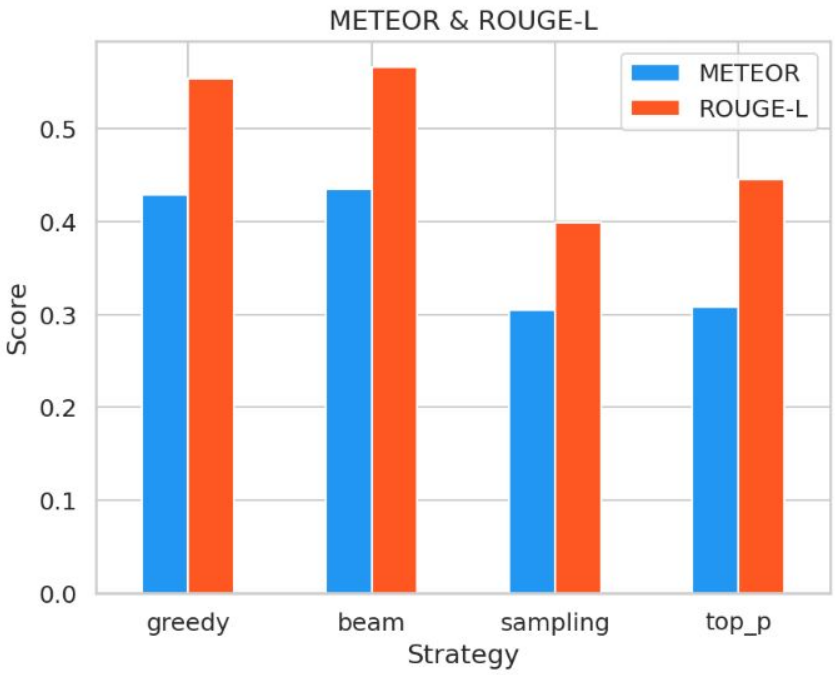
2x EPOCHS+COSINE LR

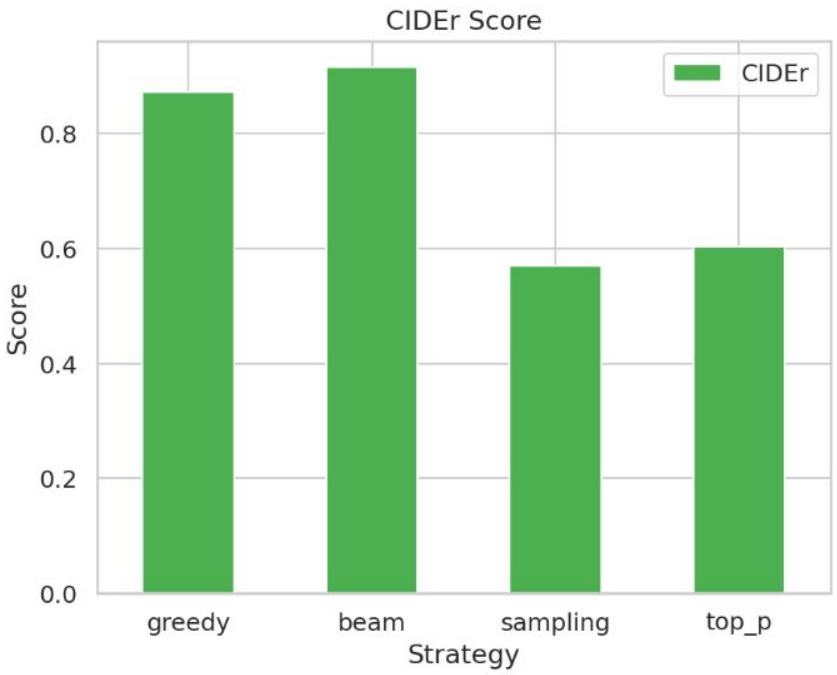




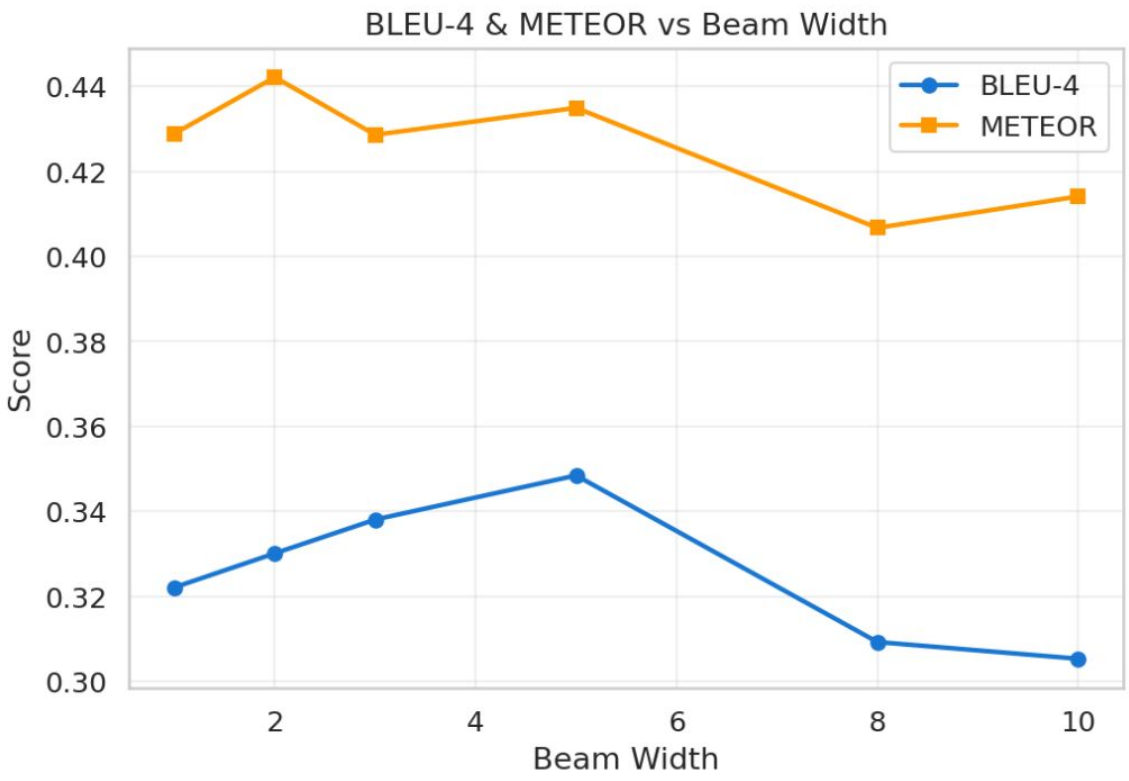
2x EPOCHS+COSINE LR

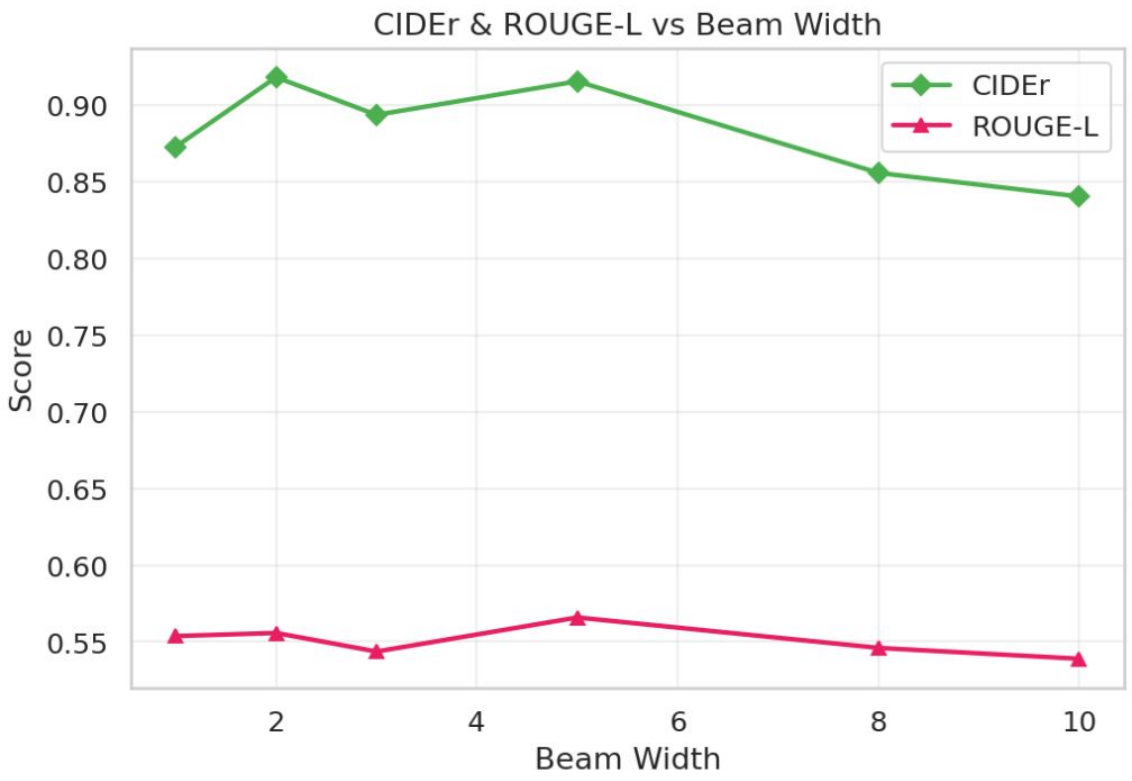




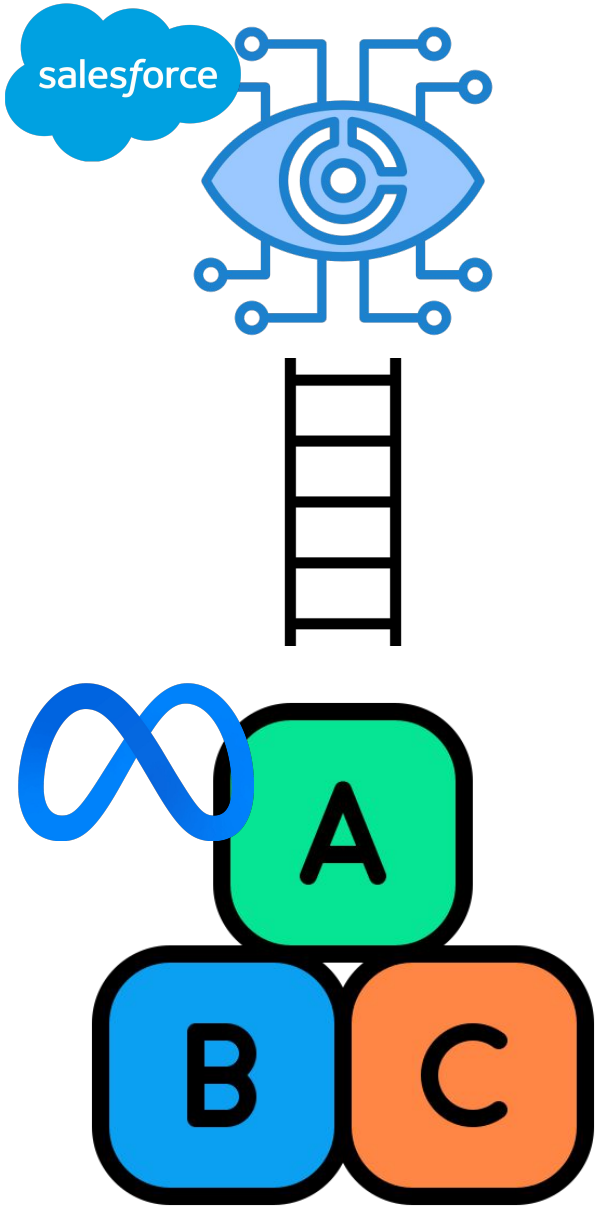


2x EPOCHS+COSINE LR





FINAL MODEL ARCHITECTURE



BLIP 2 2.7B

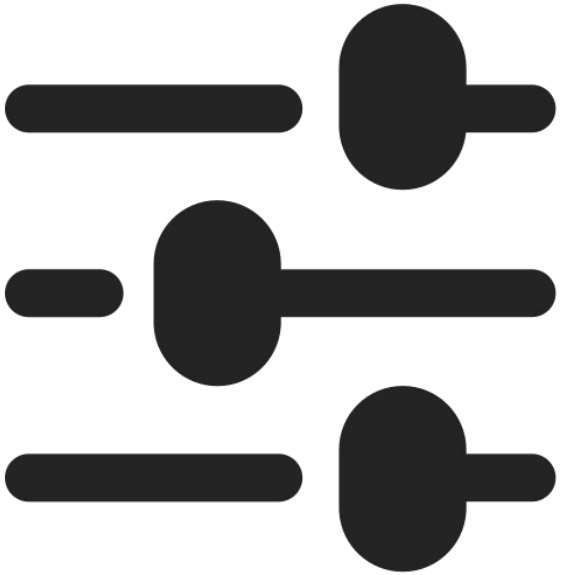
Q-Former

OPT 2.7B

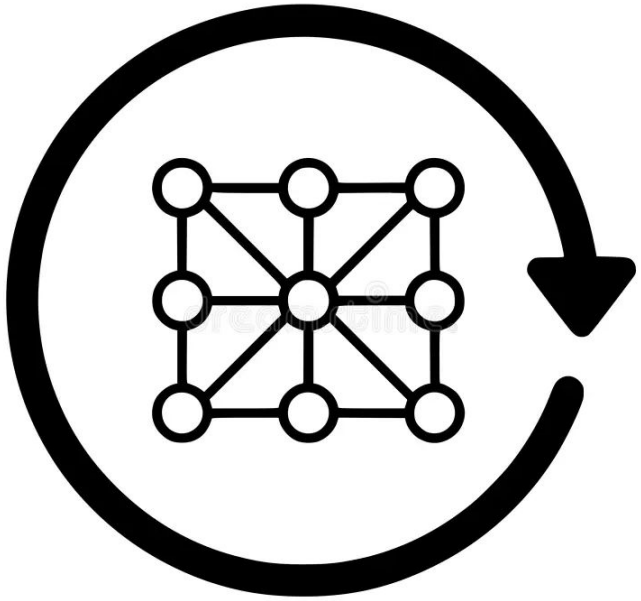
FINAL MODEL ARCHITECTURE

bfloat16

10k images

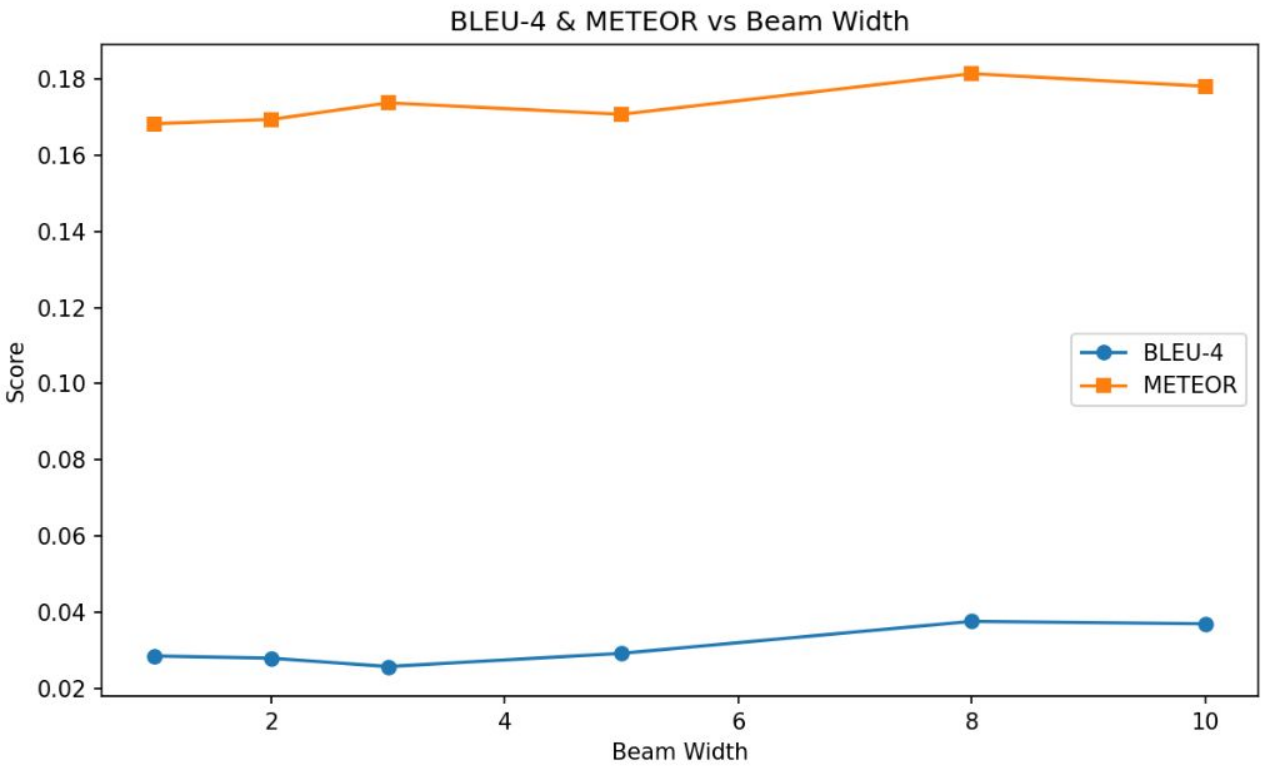


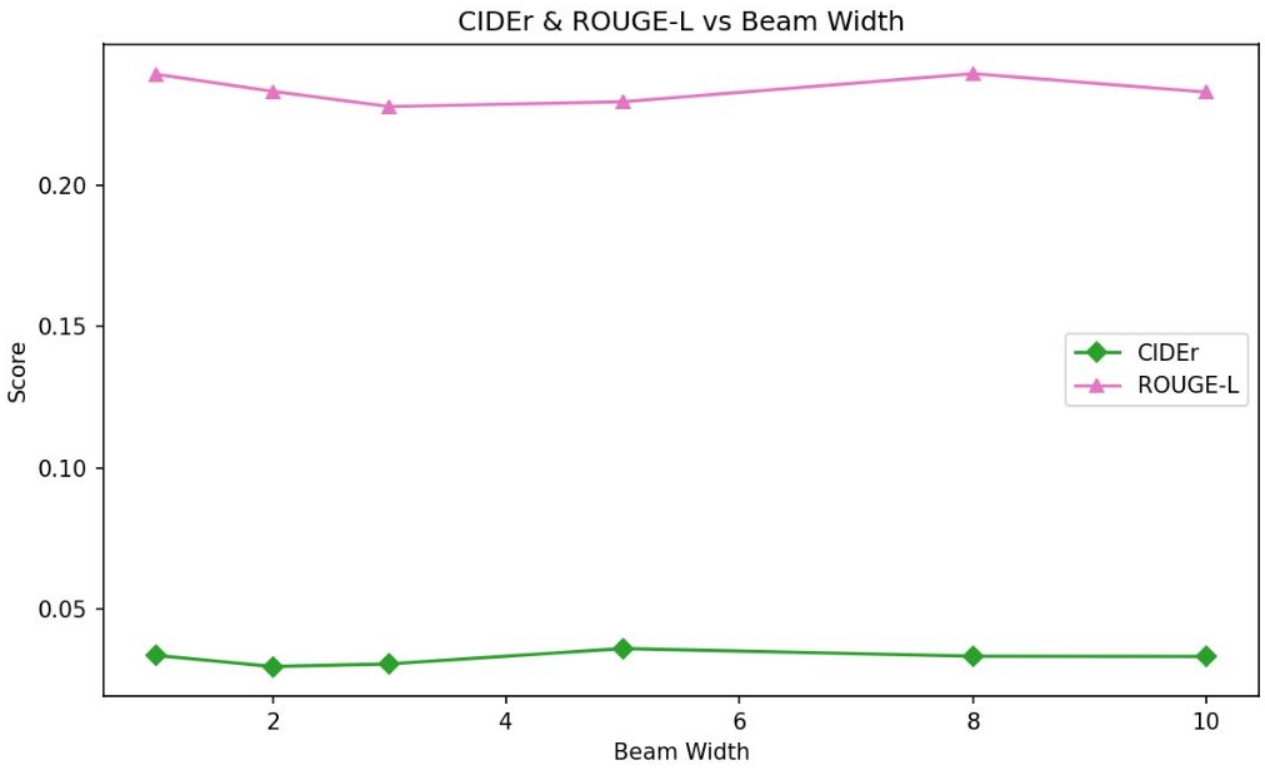
LoRA



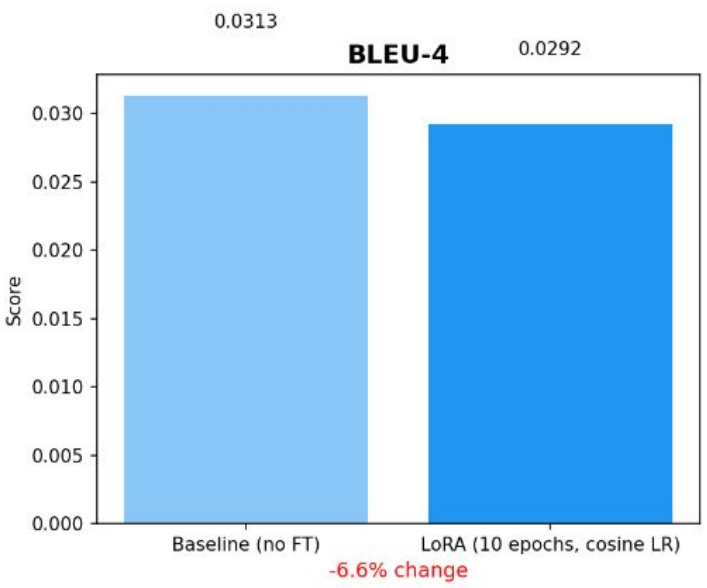
10 epochs
+ cosine LR

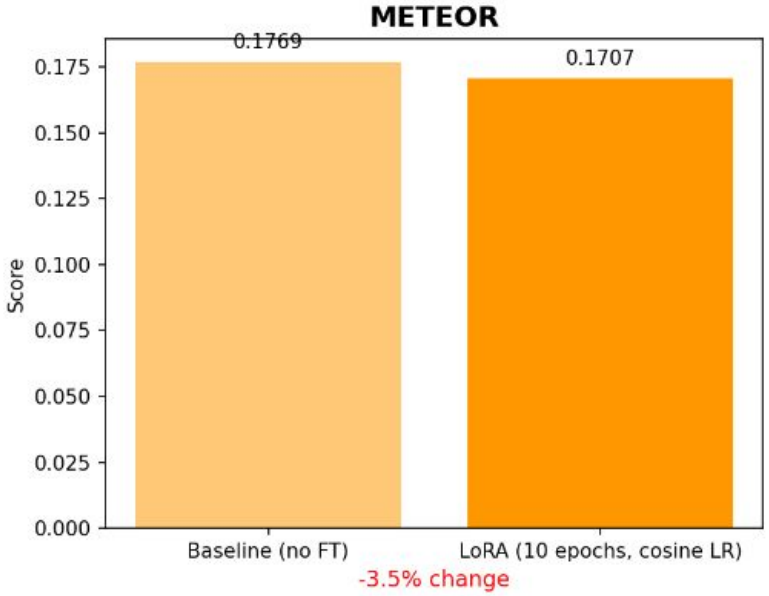
FINAL MODEL RESULTS

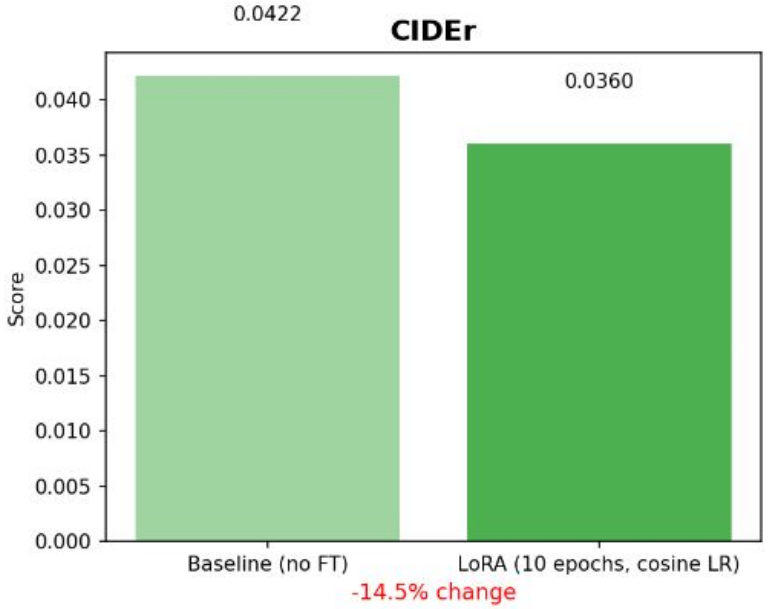


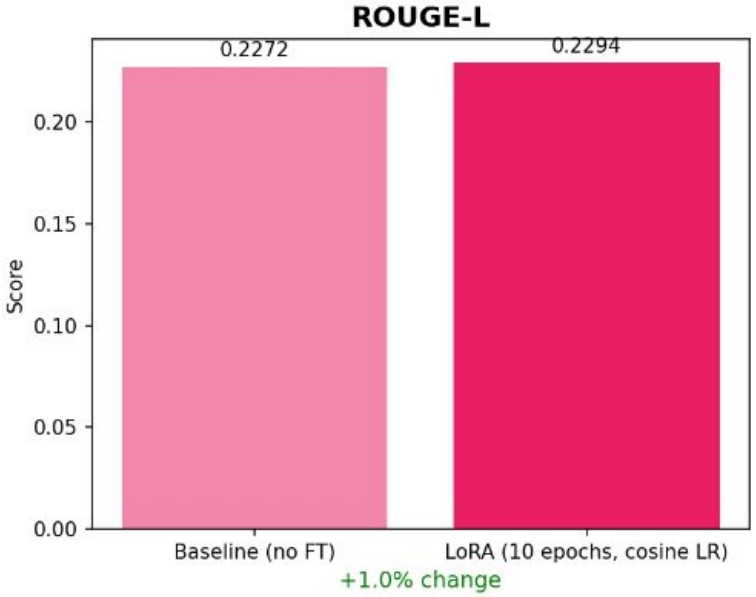


FINAL MODEL RESULTS









DEMO